

PROJECT PROPOSAL



THE LOGO DETECTION IN AN ILLEGITIMATE WEB PAGE

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**BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

MAE FAH LUANG UNIVERSITY

2022

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**A SENIOR PROJECT SUBMITTED TOMAE
FAH LUANG UNIVERSITY IN PARTIAL FULFILLMENT OF
THE REQUIREMENTS FOR THE DEGREE OF
BACHELOR OF ENGINEERING
IN COMPUTER ENGINEERING**

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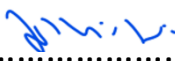
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Abstract

This senior project. We will present to you the creation and development of an application Detect the Mae Fah Luang University logo. Used to check the Mae Fah Luang University logo on dark website around the world by use machine learning to training AI to do image processing logos university to detect images and retrieve the domain of dark websites by “WHO IS” and use an application to detect the Mae Fah Luang University logo on the dark website.

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CHAPTER 1

INTRODUCTION

1.1 Background and Rationale

Logos that we saw today. We can't know if a logo is copyrighted or allowed to be used or traded on the dark web, so we must make this AI to detect misuse. And scan to detect websites that use that logo to check if it's true or not.

However, it is important to note that using the dark web can use Mah Fah Luang university logo. Using it in an illegal way may cause the university to discredit the reputation. We therefore develop an AI to detect the university logo on dark websites whether it is a proper logo or not.

1.2 Objective

1.2.1 To detect Mae Fah Luang University's logo being used illegally.

1.3 Scope

1.3.1 Display Mae Fah Luang University logo on the dark web.

1.4 Methodology

1.4.1 Preparation and project design

1.4.2 Proposal Evaluation

1.4.3 Prepare Knowledge

1.4.4 Develop AI

1.4.5 Presentation & Documentation

1.4.6 System development

1.4.7 Application testing

1.4.8 Documentation / Report

1.4.9 Final Project presentation

1.5 Plan

Table 1.5.1 Working plan

	Senior project 1					Senior project 2				
Plan/Month	Jan	Feb	Mar	Apr	May	Aug	Sep	Oct	Nov	Dec
Proposal preparation and defense										
Proposal Evaluation										
Prepare Knowledge										
Develop AI										
Presentation & Documentation										
System Development										
System testing										
Documentation/Report										
Final project presentation										

1.6 Expected Result

1.6.1 Found all Mae Fah Luang University logos on the dark web.

1.7 Place and Equipment

1.7.1 Place (Locate the working place such as MFU and/or other places for getting requirements, testing and installing the system)

1.7.2 Software

- Docker Desktop
- Convolutional Neural Networks (CNNs)
- VMware
- Jupyter

CHAPTER 2

LITERATURE REVIEW

2.1 Related Theory

Image processing is the process of transforming an image into a digital form and performing certain operations to get some useful information from it. The image processing system usually treats all images as 2D signals when applying certain predetermined signal processing methods.

Types of Image Processing

There are five main types of image processing:

Visualization - Find objects that are not visible in the image.

Recognition - Distinguish or detect objects in the image.

Sharpening and restoration - Create an enhanced image from the original image.

Pattern recognition - Measure the various patterns around the objects in the image.

Retrieval - Browse and search for images from a large database of digital images that are like the original image.

Fundamental Image Processing

Sampling and Quantization: Sampling refers to the process of converting a continuous image into a discrete representation by selecting specific points (pixels) from the continuous domain. Quantization involves assigning discrete intensity levels or color values to each pixel based on the continuous range of values present in the image.

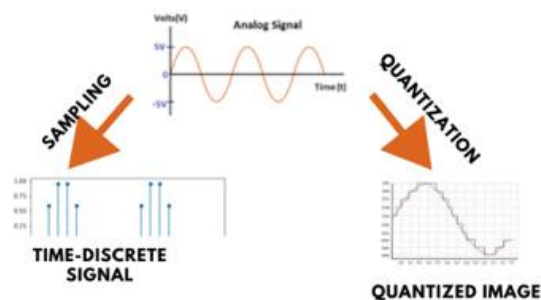


Figure 2.1.1 Fundamental Image Processing

Image Enhancement: Image enhancement techniques aim to improve the visual quality or interpretability of an image. This can involve operations like contrast adjustment, brightness correction, histogram equalization, and spatial filtering to enhance edges or details.

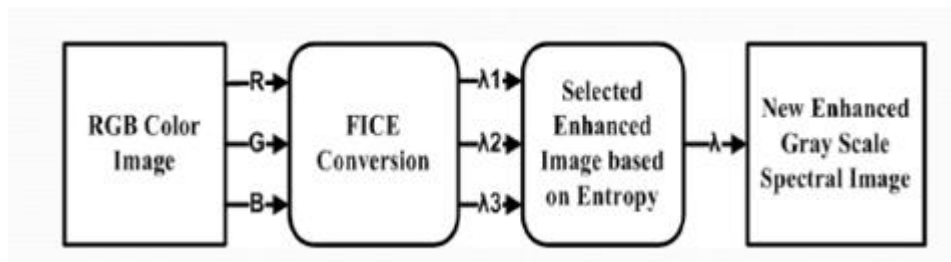


Figure 2.1.2 Block diagram of proposed Image Enhancement algorithm

Image Filtering: Image filtering involves applying various filters to an image to enhance or modify specific characteristics. Common filters include Gaussian, median, and bilateral filters, which are used for noise reduction, smoothing, and edge detection.

Image Restoration: Image restoration techniques aim to recover or improve the quality of images that are degraded by noise, blurring, or other artifacts. Methods such as denoising filters (e.g., Wiener filter, Median filter), deblurring algorithms, and inpainting techniques are commonly used for image restoration.

Image Segmentation: Image segmentation involves dividing an image into meaningful regions or segments. It is used for tasks like object recognition, boundary detection, and image understanding. Techniques such as thresholding, region-based segmentation, and clustering algorithms are commonly employed.



Figure 2.1.3 Image Segmentation

Image Compression: Image compression techniques reduce the size of an image while preserving its visual quality. Compression methods can be categorized as either lossless (e.g., Run-Length Encoding, Huffman Coding) or lossy (e.g., Discrete Cosine Transform, Wavelet Transform). Compression algorithms like JPEG and PNG are widely used in image storage and transmission.

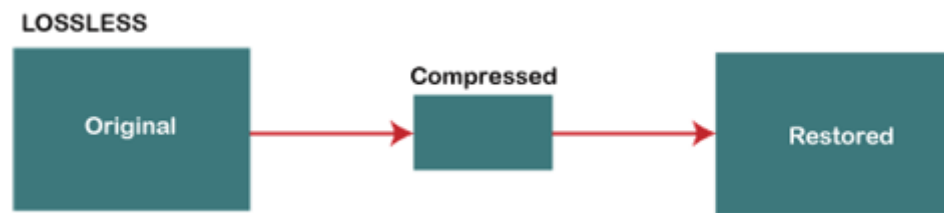


Figure 2.1.4 Image Compression

Image Registration: Image registration is the process of aligning or matching two or more images taken at different times, from different viewpoints, or with different modalities. It is used in applications like medical image analysis, remote sensing, and augmented reality. Techniques for image registration include intensity-based methods, feature-based methods, and mutual information-based methods.



Figure 2.1.5 Image Registration

Morphological Operations: Morphological operations are a set of operations used for analyzing and manipulating image shapes. They include dilation, erosion, opening, closing, and morphological gradient. These operations are used for tasks such as noise removal, object extraction, and boundary detection.

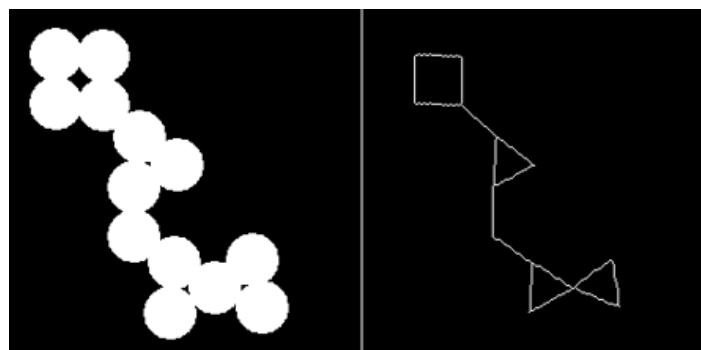
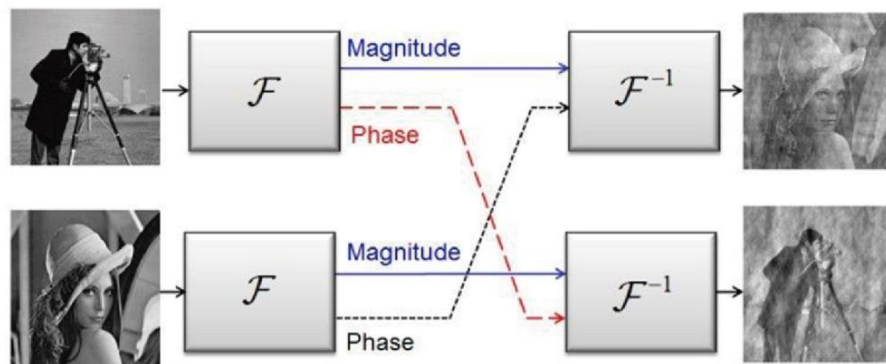


Figure 2.1.6 Morphological Operations

Importance of Phase in Image Processing

Phase, in a nutshell, contains information about the positions of features. Phase-only and magnitude-only photos cannot be combined to produce the original. To obtain the original, multiply them in the Fourier domain and reverse the transformation. A repeated waveform's phase describes the position or timing of a particular point within a wave cycle. Instead of the actual absolute phases of the signals, the phase difference between waves usually matters.



Ringling Effect in Image Processing

The unpleasant ringing effect often referred to as the Gibbs phenomenon in mathematical approaches to image processing, is an artifact that appears as rippling ripples close to sharp edges in photos and videos. The loss or distortion of high-frequency information in the image results in this effect.

Applications of Image Processing

- Medical Image Retrieval

Image processing has been extensively used in medical research and has enabled more efficient and accurate treatment plans. For example, it can be used for the early detection of breast cancer using a sophisticated nodule detection algorithm in breast scans. Since medical usage calls for highly trained image processors, these applications require significant implementation and evaluation before they can be accepted for use.

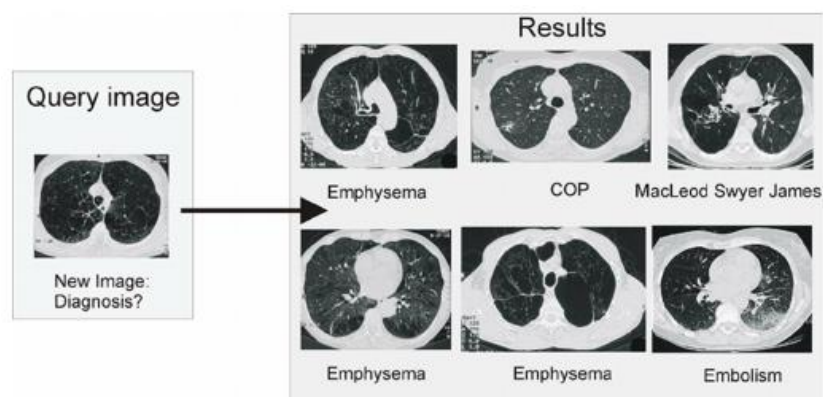


Figure 2.1.7 Medical Image Retrieval

- Face Detection

One of the most common applications of image processing that we use today is face detection. It follows deep learning algorithms where the machine is first trained with the specific features of human faces, such as the shape of the face, the distance between the eyes, etc. After teaching the machine these human face features, it will start to accept all objects in an image that resemble a human face. Face detection is a vital tool used in security, biometrics and even filters available on most social media apps these days.



Figure 2.1.8 Face Detection

2.2 Related Research

- Currently artificial intelligence is one very important technological tool widely used in modern society. Particularly, Deep Learning (DL) has several applications due to its ability to learn robust representations from images. Convolutional Neural Networks (CNN) is the main DL architecture for image classification. Based on the great attention that CNNs have had in the last years, we present a review of the use of CNN applied to different automatic processing tasks of fruit images: classification, quality control, and detection. We observe that in the last two years (2019–2020), the use of CNN for fruit recognition has greatly increased obtaining excellent results, either by using new models or with pre-trained networks for transfer learning. It is worth noting that different types of images are used in datasets according to the task performed. Besides, this article presents the fundamentals, tools, and two examples of the use of CNNs for fruit sorting and quality control.

[Additional Information:

<https://www.mdpi.com/2076-3417/10/10/3443?fbclid=IwAR0MICNZBXDFY4rvhzTiEvjOnF6EK9oIOaBcO-oC1ob4hfQC-UAn40VAdZk>]

- Deep Learning algorithms are designed in such a way that they mimic the function of the human cerebral cortex. These algorithms are representations of deep neural networks i.e., neural networks with many hidden layers. Convolutional neural networks are deep learning algorithms that can train large datasets with millions of parameters, in the form of 2D images as input and convolve it with filters to produce the desired outputs. In this article, CNN models are built to evaluate its performance on image recognition and detection datasets. The algorithm is implemented on MNIST and CIFAR-10 dataset and its performance are evaluated. The accuracy of models on MNIST is 99.6 %, CIFAR-10 is using real-time data augmentation and dropout on CPU units.

[Additional Information:

https://www.researchgate.net/publication/332826568_Convolutional_Neural_Network_CNN_for_Image_Detection_and_Recognition]

2.3 Related Technology

2.3.1 Docker Desktop

Docker is a software platform that allows you to build, test, and deploy applications quickly. Docker packages software into standardized units called containers that have everything the software needs to run including libraries, system tools, code, and runtime. Using Docker, you can quickly deploy and scale applications into any environment and know your code will run.



Figure 2.3.1 Docker Desktop

2.3.2 Jupyter

Project Jupyter is a non-profit, open-source project, born out of the IPython Project in 2014 as it evolved to support interactive data science and scientific computing across all programming languages.



Figure 2.3.2 Jupyter

2.3.3 Convolutional Neural Networks (CNNs)

Convolutional Neural Networks (CNNs) are biologically inspired variants of MLPs. From Hubel and Wiesel's early work on the cat's visual cortex, we know the visual cortex contains a complex arrangement of cells. These cells are sensitive to small sub-regions of the visual field, called a receptive field. The sub-regions are tiled to cover the entire visual field. These cells act as local filters over the input space and are well-suited to exploit the strong spatially local correlation present in natural images.

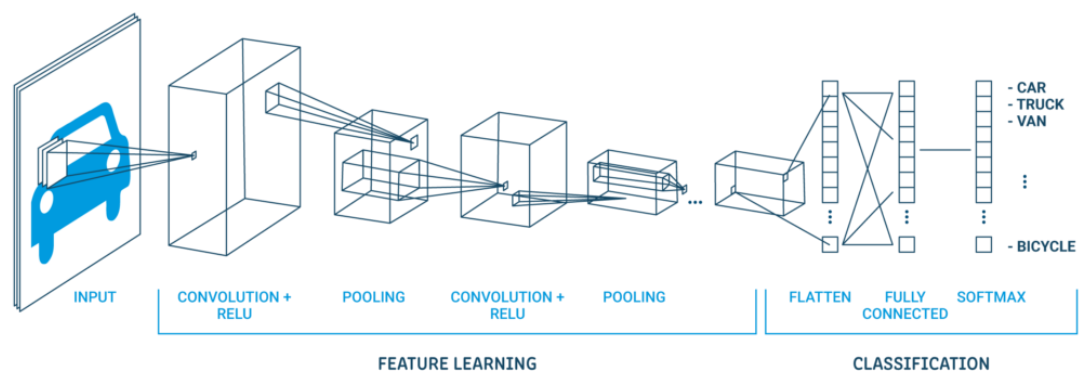


Figure 2.3.3 An Example Convolutional Neural Networks (CNNs)

2.3.4 OpenCV

OpenCV (Open-Source Computer Vision Library) is an open source computer vision and machine learning software library. OpenCV was built to provide a common infrastructure for computer vision applications and to accelerate the use of machine perception in the commercial products. Being an Apache 2 licensed product, OpenCV makes it easy for businesses to utilize and modify the code.



Figure 2.3.4 OpenCV

2.3.5 Python

Python is a popular programming language. It was created by Guido van Rossum and released in 1991.

It is used for:

- web development (server-side),
- software development,
- mathematics,
- system scripting.

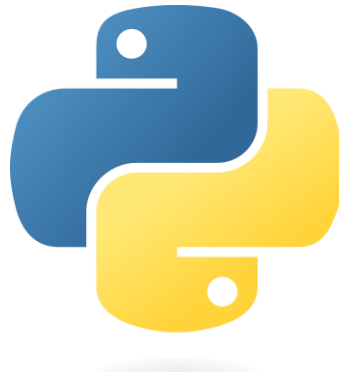


Figure 2.3.5 Python

2.3.6 Deep Learning Framework

Deep learning (DL) frameworks offer building blocks for designing, training, and validating deep neural networks through a high-level programming interface. Widely-used DL frameworks, such as PyTorch, TensorFlow, PyTorch Geometric, DGL, and others, rely on GPU-accelerated libraries, such as cuDNN, NCCL, and DALI to deliver high-performance, multi-GPU-accelerated training.

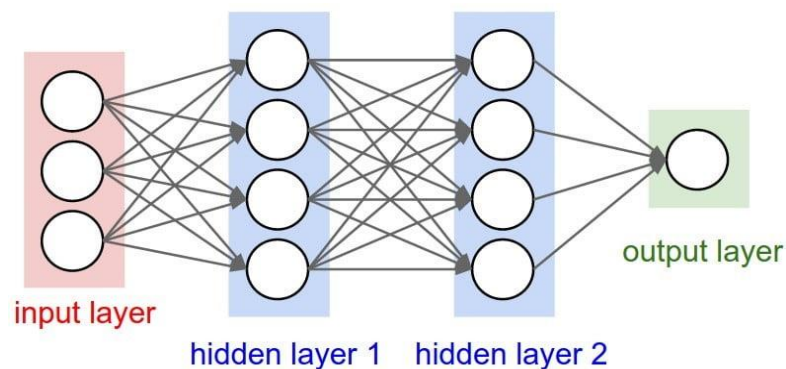


Figure 2.3.6 Deep Learning

2.3.7 Image Filtering

Image filtering is changing the appearance of an image by altering the colors of the pixels. Increasing the contrast as well as adding a variety of special effects to images are some of the results of applying filters. To obtain a high success rate of OCR (optical character recognition) performed on text images, the main target of filters is, however, to reduce the noise around characters in the image.

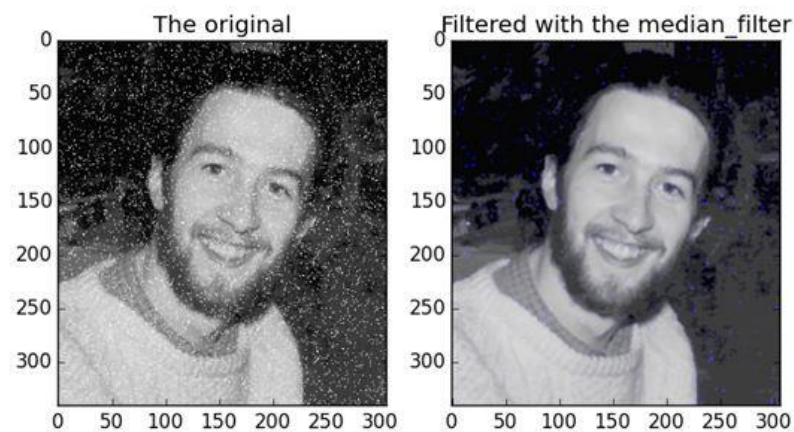


Figure 2.3.7 Image Filtering

CHAPTER 3

ANALYSIS AND DESIGN

3.1 Existing System

Face detection uses machine learning and artificial neural network technology, and plays an important role in face tracking, face analysis and facial recognition. In face analysis, face detection uses facial expressions to identify which parts of an image or video should be focused on to determine age, gender and emotions. In a facial recognition system. Face Detection could be demonstrated by a use case diagram in Figure 3.1

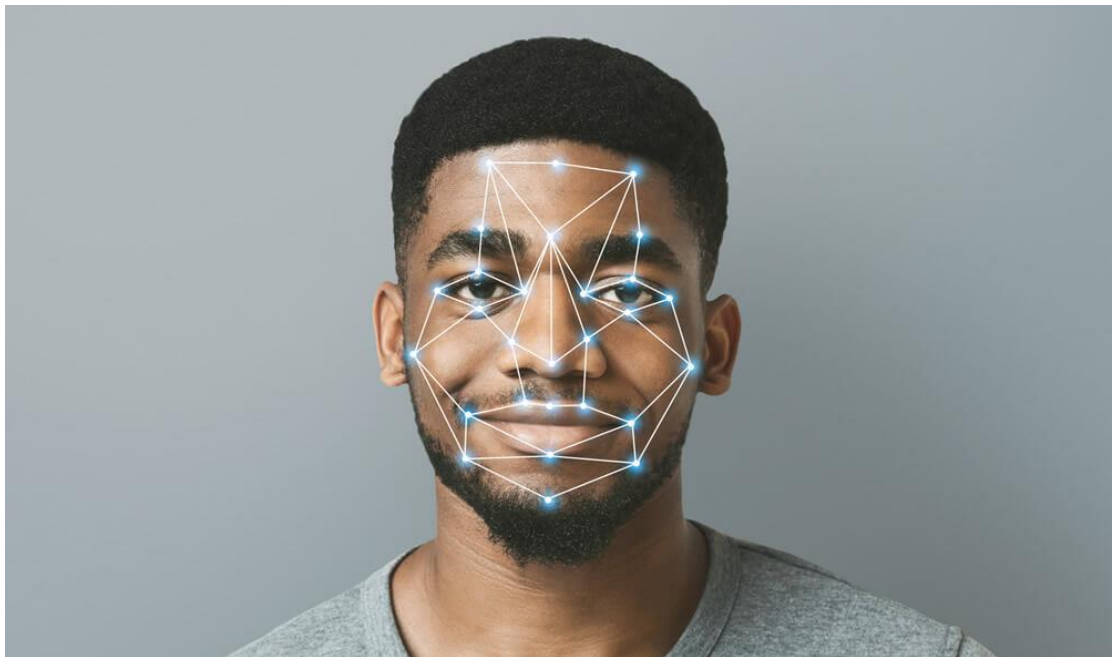


Figure 3.1.1 Face Detection

3.2 New System

```
[57]: yhat = model.predict(np.expand_dims(resize/255, 0))  
1/1 [=====] - 0s 58ms/step
```

```
[58]: yhat
```

```
[58]: array([[0.9996988]], dtype=float32)
```

```
[59]: if yhat > 0.5:  
      print(f'Predicted class is NotLogo')  
      else:  
      print(f'Predicted class is LogoMFU!!')
```

Predicted class is NotLogo

```
[49]: import cv2
```

```
[55]: img = cv2.imread('cattest.jpg')  
      plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))  
      plt.show()
```



Figure 3.2.1 New System

Figure 3.2.2 New System

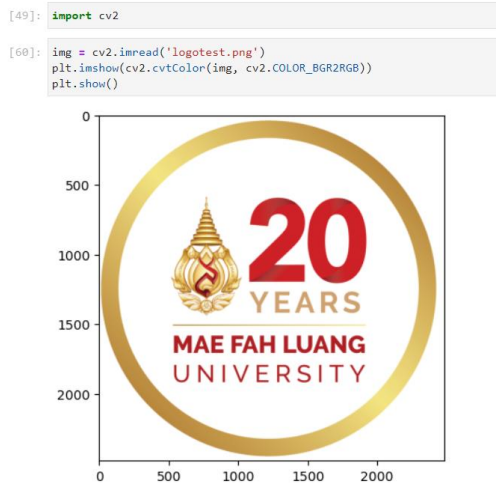


Figure 3.2.3 New System

```
[62]: yhat = model.predict(np.expand_dims(resize/255, 0))
      1/1 [=====] - 0s 62ms/step
```

```
[63]: yhat
```

```
[63]: array([[0.00149947]], dtype=float32)
```

```
[64]: if yhat > 0.5:
      print(f'Predicted class is NotLogo')
      else:
      print(f'Predicted class is LogoMFU!!')
```

Predicted class is LogoMFU!!

Figure 3.2.4 New System

The system can predict the picture as a number and check if the picture is the university's logo or not.

3.3 Activity diagram

Analysis of a new system derives an activity diagram as shown in Figure 3.3.1

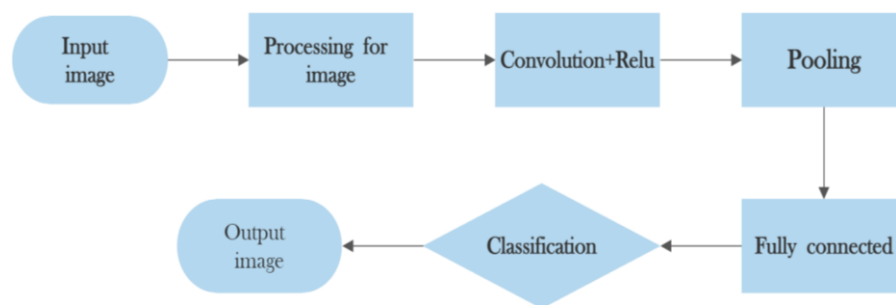


Figure 3.3.1 Activity diagram of the new system

CHAPTER 4

CONCLUSION

4.1 Conclusion

We have set the scope of Artificial Intelligence development from research planning and found the related research for reference from analysis design and system design attractive and usable and more efficient. We set the Artificial Intelligence development for a main role:

- Develop AI System

4.2 Discussion

This is new to us. which must start learning and starting over therefore taking longer than usual including study, research, data acquisition and experimentation and checking reference research details and verify the accuracy of the information.

As a result, it was successful in developing AI, but the logo of the university could not be found.

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